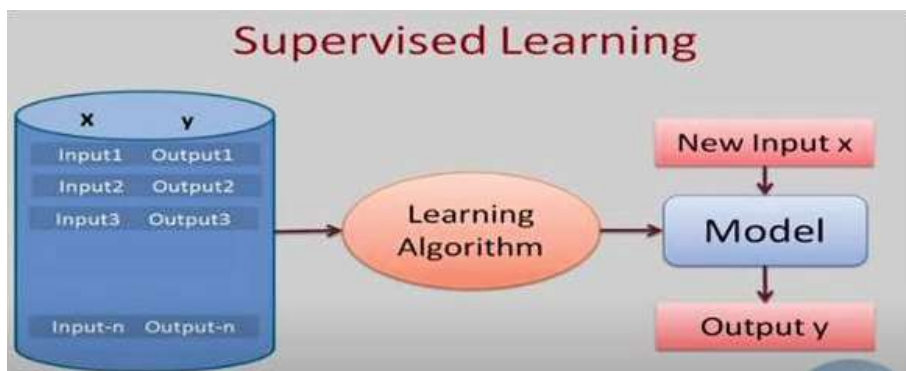


Machine Learning

Types of Learning

1. Supervised Machine Learning

- Supervised learning is one of the most basic types of machine learning.
- In this type, the machine learning algorithm is trained on labeled data.
- Even though the data needs to be labeled accurately for this method to work, supervised learning is extremely powerful when used in the right circumstances.



Machine Learning

Types of Learning

1. Supervised Machine Learning

- In supervised learning, the ML algorithm is given a small training dataset to work with.
- This training dataset is a smaller part of the bigger dataset and serves to give the algorithm a basic idea of the problem, solution, and data points to be dealt with.
- The training dataset is also very similar to the final dataset in its characteristics and provides the algorithm with the labeled parameters required for the problem.
- The algorithm then finds relationships between the parameters given, essentially establishing a cause and effect relationship between the variables in the dataset.
- At the end of the training, the algorithm has an idea of how the data works and the relationship between the input and the output.

Machine Learning

Types of Learning

1. Supervised Machine Learning

- In supervised learning, learning data comes with description, labels, targets or desired outputs and the objective is to find a general rule that maps inputs to outputs.
- This kind of learning data is called labeled data.
- The learned rule is then used to label new data with unknown outputs.
- Supervised learning involves building a machine learning model that is based on labeled samples.
- For example, if we build a system to estimate the price of a plot of land or a house based on various features, such as size, location, and so on, we first need to create a database and label it. We need to teach the algorithm what features correspond to what prices. Based on this data, the algorithm will learn how to calculate the price of real estate using the values of the input features.

Machine Learning

Types of Learning

1. Supervised Machine Learning

- Supervised learning is commonly used in real world applications, such as face and speech recognition, products or movie recommendations, and sales forecasting.
- Supervised learning deals with learning a function from available training data. Here, a learning algorithm analyzes the training data and produces a derived function that can be used for mapping new examples.
- Supervised learning can be further classified into two types - Regression and Classification.

a) Regression

- Regression trains on and predicts a continuous-valued response, for example predicting real estate prices.
- When output Y is discrete valued, it is classification and when Y is continuous, then it is Regression.

Machine Learning

Types of Learning

1. Supervised Machine Learning

- Regression algorithms are used if there is a relationship between the input variable and the output variable.
- It is used for the prediction of continuous variables, such as Weather forecasting, Market Trends, etc.

i) Linear Regression

ii) Regression Trees

iii) Non-Linear Regression

iv) Bayesian Linear Regression

v) Polynomial Regression

vi) Logistic Regression

Machine Learning

Types of Learning

1. Supervised Machine Learning

b) Classification

- Classification attempts to find the appropriate class label, such as analyzing positive/negative sentiment, male and female persons, benign and malignant tumors, secure and unsecure loans etc.
- Classification algorithms are used when the output variable is categorical, which means there are two classes such as Yes-No, Male-Female, True-false, etc.

i) Decision Trees

ii) Random Forest

iii) Support vector Machines

iv) Neural network

Common examples of supervised learning include
classifying emails into spam and not-spam categories, labeling web pages based on their content, and voice recognition.

Machine Learning

Types of Learning

1. Unsupervised Machine Learning

- Unsupervised machine learning holds the advantage of being able to work with unlabeled data.
- This means that human labor is not required to make the dataset machine-readable, allowing much larger datasets to be worked on by the program.
- In supervised learning, the labels allow the algorithm to find the exact nature of the relationship between any two data points.
- However, unsupervised learning does not have labels to work off of, resulting in the creation of hidden structures.
- Relationships between data points are perceived by the algorithm in an abstract manner, with no input required from human beings.

Machine Learning

Types of Learning

1. Unsupervised Machine Learning

- The creation of these hidden structures is what makes unsupervised learning algorithms versatile.
- Instead of a defined and set problem statement, unsupervised learning algorithms can adapt to the data by dynamically changing hidden structures.
- This offers more post-deployment development than supervised learning algorithms.
- Unsupervised learning is used to detect anomalies, outliers, such as fraud or defective equipment, or to group customers with similar behaviours for a sales campaign.
- It is the opposite of supervised learning. There is no labeled data here.
- When learning data contains only some indications without any description or labels, it is up to the coder or to the algorithm to find the structure of the underlying data, to discover hidden patterns, or to determine how to describe the data. This kind of learning data is called unlabeled data.

Machine Learning

Types of Learning

1. Unsupervised Machine Learning

- Suppose that we have a number of data points, and we want to classify them into several groups. We may not exactly know what the criteria of classification would be.
- So, an unsupervised learning algorithm tries to classify the given dataset into a certain number of groups in an optimum way.
- Unsupervised learning algorithms are extremely powerful tools for analyzing data and for identifying patterns and trends.
- They are most commonly used for clustering similar input into logical groups.
- It has two types clustering and Association

Machine Learning

Types of Learning

2. Unsupervised Machine Learning

a. Clustering:

- Clustering is a method of grouping the objects into clusters such that objects with most similarities remains into a group and has less or no similarities with the objects of another group.
- Cluster analysis finds the commonalities between the data objects and categorizes them as per the presence and absence of those commonalities.

b. Association:

- An association rule is an unsupervised learning method which is used for finding the relationships between variables in the large database.
- It determines the set of items that occurs together in the dataset.
- Association rule makes marketing strategy more effective. Such as people who buy X item (suppose a bread) are also tend to purchase Y (Butter/Jam) item.
- A typical example of Association rule is Market Basket Analysis.

Machine Learning

2. Unsupervised Machine Learning

The list of some popular unsupervised learning algorithms:

- K-means clustering
- KNN (K-Nearest Neighbors)
- Hierarchical clustering
- Anomaly detection
- Neural Networks
- Principle Component Analysis
- Independent Component Analysis
- Apriori algorithm
- Singular value decomposition

Machine Learning

Sr.No.	Supervised	Unsupervised
1	Supervised learning algorithms are trained using labeled data.	Unsupervised learning algorithms are trained using unlabeled data.
2	Supervised learning model takes direct feedback to check if it is predicting correct output or not.	Unsupervised learning model does not take any feedback.
3	Supervised learning model predicts the output.	Unsupervised learning model finds the hidden patterns in data.
4	In supervised learning, input data is provided to the model along with the output.	In unsupervised learning, only input data is provided to the model.
5	The goal of supervised learning is to train the model so that it can predict the output when it is given new data.	The goal of unsupervised learning is to find the hidden patterns and useful insights from the unknown dataset.

Machine Learning

Sr.No.	Supervised	Unsupervised
1	Supervised learning requires a target variable to train the model.	Unsupervised learning does not require a target variable to train the model.
2	Supervised learning can be categorized in Classification and Regression problems.	Unsupervised Learning can be categorized in Clustering and Associations problems.
3	Supervised learning model produces an accurate result.	Unsupervised learning model may give less accurate result as compared to supervised learning.
4	Supervised learning is more close to the human learning process.	Unsupervised learning is more close to the human learning process.

Machine Learning

Types of Learning

3.Semi Supervised Machine Learning

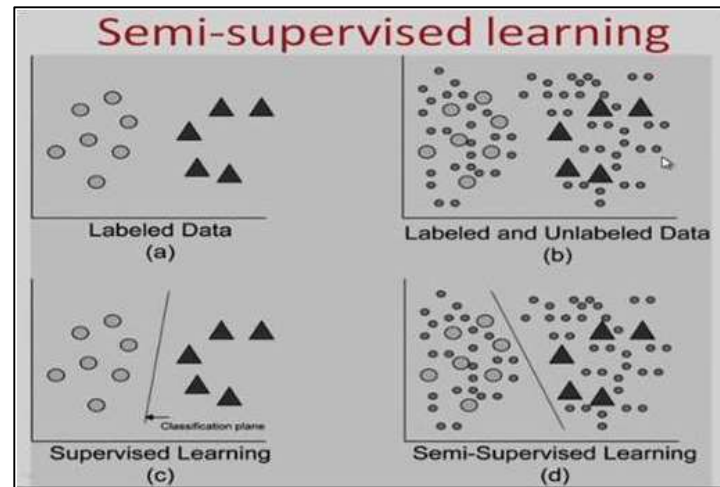
- The most basic disadvantage of any Supervised Learning algorithm is that the dataset has to be hand-labeled either by a Machine Learning Engineer or a Data Scientist. This is a very costly process, especially when dealing with large volumes of data. The most basic disadvantage of any Unsupervised Learning is that its application spectrum is limited.
- To counter these disadvantages, the concept of Semi-Supervised Learning was introduced.
- It is partly supervised and partly unsupervised .
- If some learning samples are labeled, but some other are not labeled, then it is semi-supervised learning.
- It makes use of a large amount of unlabeled data for training and a small amount of labeled data for testing.

Machine Learning

Types of Learning

3. Semi Supervised Machine Learning

- Semi-supervised learning is applied in cases where it is expensive to acquire a fully labeled dataset while more practical to label a small subset.
- Supervised learning: where a student is under the supervision of a teacher at both home and school, Unsupervised learning: where a student has to figure out a concept himself and Semi-Supervised learning: where a teacher teaches a few concepts in class and gives questions as homework which are based on similar concepts.



Machine Learning

Types of Learning

4. Reinforcement Learning

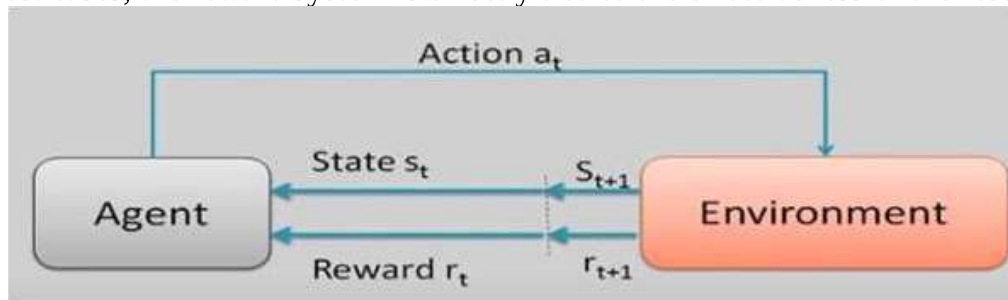
- Reinforcement learning directly takes inspiration from how human beings learn from data in their lives.
- It features an algorithm that improves upon itself and learns from new situations using a trial-and-error method.
- Favourable outputs are encouraged or 'reinforced', and non-favourable outputs are discouraged or 'punished'.
- Based on the psychological concept of conditioning, reinforcement learning works by putting the algorithm in a work environment with an interpreter and a reward system.
- In every iteration of the algorithm, the output result is given to the interpreter, which decides whether the outcome is favourable or not.

Machine Learning

Types of Learning

4. Reinforcement Learning

- In case of the program finding the correct solution, the interpreter reinforces the solution by providing a reward to the algorithm.
- If the outcome is not favourable, the algorithm is forced to reiterate until it finds a better result.
- In most cases, the reward system is directly tied to the effectiveness of the result.



Machine Learning

Types of Learning

4. Reinforcement Learning

- In typical reinforcement learning use-cases, such as finding the shortest route between two points on a map, the solution is not an absolute value.
- Instead, it takes on a score of effectiveness, expressed in a percentage value.
- The higher this percentage value is, the more reward is given to the algorithm.
- Thus, the program is trained to give the best possible solution for the best possible reward.
- Here learning data gives feedback so that the system adjusts to dynamic conditions in order to achieve a certain objective.
- The system evaluates its performance based on the feedback responses and reacts accordingly.
- The best known instances include self-driving cars and chess master algorithm AlphaGo.
- There are two important learning models in reinforcement learning: Markov Decision Process & Q learning

Machine Learning

Types of Learning

